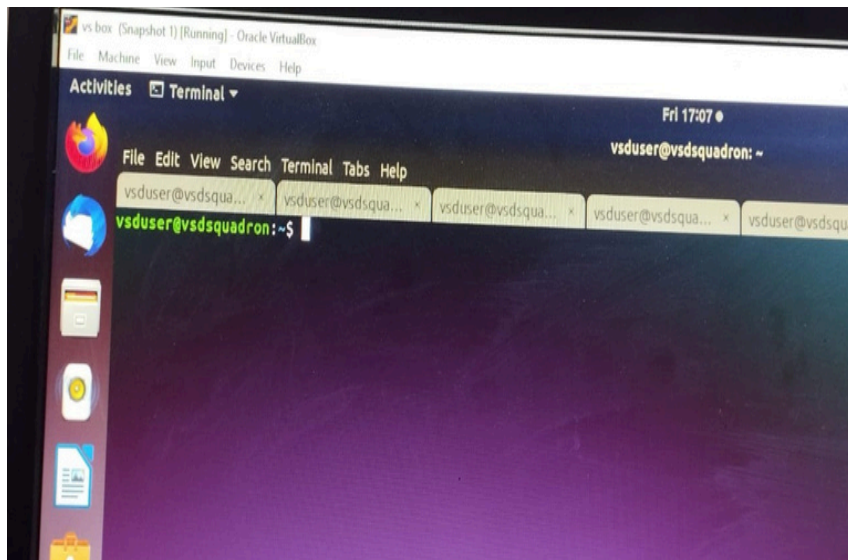


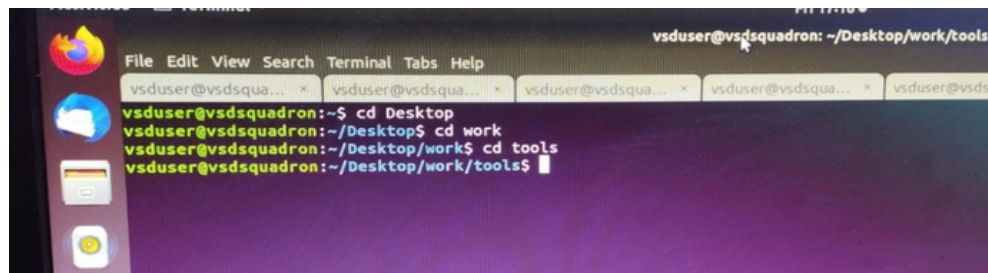
OpenLANE Directory Structure

The aim of openlane is to automate the entire RTL to GSI flow and make it clean and opensource without human intervention.

Opening the virtual machine tool

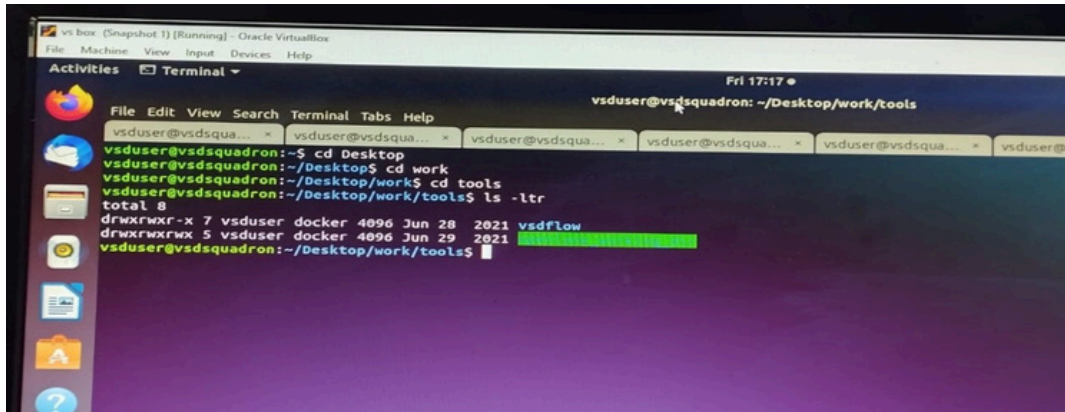


Reach out to “tools” directory through “work “ directory in the “desktop “



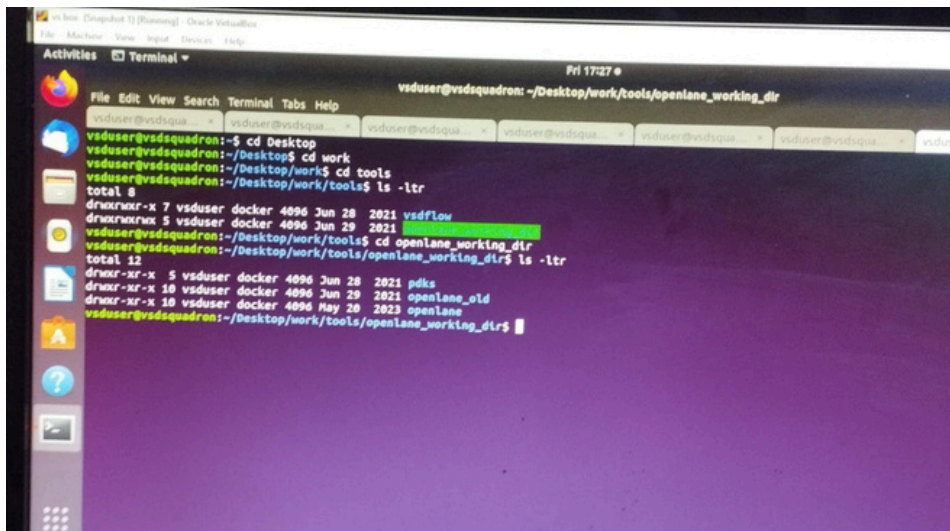
list the contents of the directory “ tools”

CHIP DESIGNING COURSE



```
vs box (Snapshot 1) [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Activities Terminal v
vdsuser@vdsquadron: ~/Desktop/work/tools
vdsuser@vdsquadron:~$ cd Desktop
vdsuser@vdsquadron:~/Desktop$ cd work
vdsuser@vdsquadron:~/Desktop/work$ cd tools
vdsuser@vdsquadron:~/Desktop/work/tools$ ls -ltr
total 8
drwxrwxr-x 7 vdsuser docker 4096 Jun 28 2021 vsdflow
drwxrwxrwx 5 vdsuser docker 4096 Jun 29 2021 vsdflow
vdsuser@vdsquadron:~/Desktop/work/tools$
```

Since we will be using open lane in this project we change the directory to openlane_working_dir using `cd openlane_working_dir`. Then lets list its contents using `ls -ltr`.

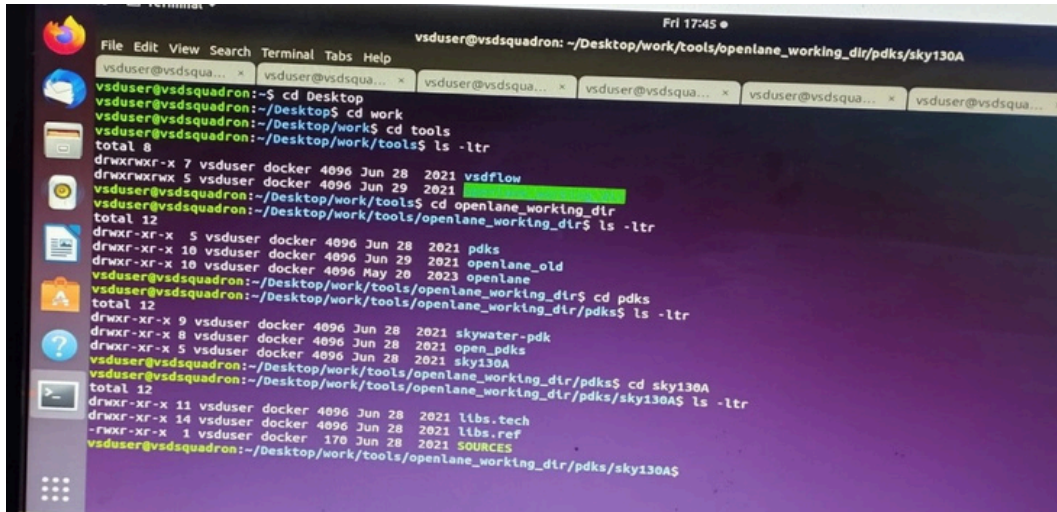


```
vs box (Snapshot 1) [Running] - Oracle VM VirtualBox
File Edit View Search Terminal Tabs Help
vdsuser@vdsquadron: ~/Desktop/work/tools/openlane_working_dir
vdsuser@vdsquadron:~$ cd Desktop
vdsuser@vdsquadron:~/Desktop$ cd work
vdsuser@vdsquadron:~/Desktop/work$ cd tools
vdsuser@vdsquadron:~/Desktop/work/tools$ ls -ltr
total 8
drwxrwxr-x 7 vdsuser docker 4096 Jun 28 2021 vsdflow
drwxrwxrwx 5 vdsuser docker 4096 Jun 29 2021 vsdflow
vdsuser@vdsquadron:~/Desktop/work/tools$ cd openlane_working_dir
vdsuser@vdsquadron:~/Desktop/work/tools/openlane_working_dir$ ls -ltr
total 12
drwxr-xr-x 5 vdsuser docker 4096 Jun 28 2021 pdks
drwxr-xr-x 10 vdsuser docker 4096 Jun 29 2021 openlane_old
drwxr-xr-x 10 vdsuser docker 4096 May 28 2023 openlane
vdsuser@vdsquadron:~/Desktop/work/tools/openlane_working_dir$
```

listing the openlane working directory we got directories pdks and openlane directory. We will be Exploring these directories one by one.

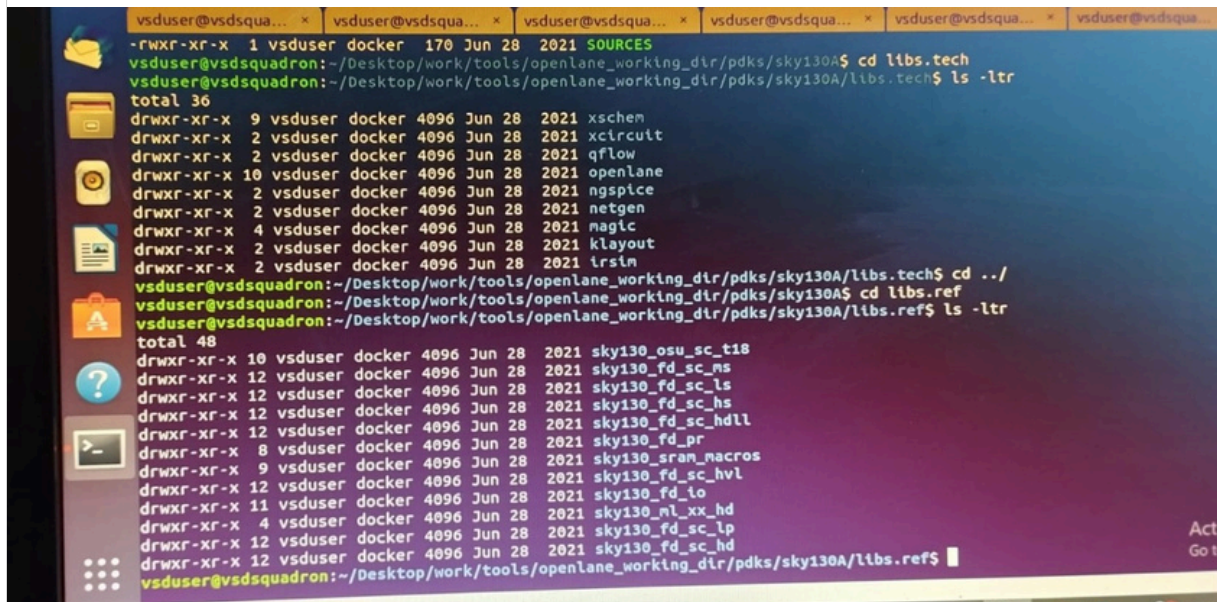
First lets explore pdks change directory by `cd pdks` and list its contents using `ls -ltr`. In the pdks to explore **sky130A**, so similarly change directory and then view contents. we observe two directories - **libs.tech** and **libs.rif**

CHIP DESIGNING COURSE



```
vsduser@vdsquadron: ~/Desktop/work/tools/openlane_working_dir/pdks/sky130A
vsduser@vdsquadron:~$ cd Desktop
vsduser@vdsquadron:~/Desktop$ cd work
vsduser@vdsquadron:~/Desktop/work$ cd tools
vsduser@vdsquadron:~/Desktop/work/tools$ ls -ltr
total 8
drwxr-xr-x 7 vsduser docker 4096 Jun 28 2021 vsdflow
drwxr-xr-x 5 vsduser docker 4096 Jun 29 2021 
vsduser@vdsquadron:~/Desktop/work/tools$ cd openlane_working_dir
vsduser@vdsquadron:~/Desktop/work/tools/openlane_working_dir$ ls -ltr
total 12
drwxr-xr-x 5 vsduser docker 4096 Jun 28 2021 pdks
drwxr-xr-x 10 vsduser docker 4096 Jun 29 2021 openlane_old
drwxr-xr-x 10 vsduser docker 4096 May 20 2023 openlane
vsduser@vdsquadron:~/Desktop/work/tools/openlane_working_dir$ cd pdks
vsduser@vdsquadron:~/Desktop/work/tools/openlane_working_dir/pdks$ ls -ltr
total 12
drwxr-xr-x 9 vsduser docker 4096 Jun 28 2021 skywater-pdk
drwxr-xr-x 8 vsduser docker 4096 Jun 28 2021 open_pdks
vsduser@vdsquadron:~/Desktop/work/tools/openlane_working_dir/pdks$ cd sky130A
vsduser@vdsquadron:~/Desktop/work/tools/openlane_working_dir/pdks/sky130A$ ls -ltr
total 12
drwxr-xr-x 11 vsduser docker 4096 Jun 28 2021 libs.tech
drwxr-xr-x 14 vsduser docker 4096 Jun 28 2021 libs.ref
-rwxr-xr-x 1 vsduser docker 170 Jun 28 2021 SOURCES
vsduser@vdsquadron:~/Desktop/work/tools/openlane_working_dir/pdks/sky130A$
```

Opening libs.ref (specific to the technology) and opening libs.tech (tool specific files) and listing its contents .(cd ../ brings out of the current directory)



```
vsduser@vdsquadron:~/Desktop/work/tools/openlane_working_dir/pdks/sky130A$ cd libs.tech
vsduser@vdsquadron:~/Desktop/work/tools/openlane_working_dir/pdks/sky130A/libs.tech$ ls -ltr
total 36
drwxr-xr-x 9 vsduser docker 4096 Jun 28 2021 xschem
drwxr-xr-x 2 vsduser docker 4096 Jun 28 2021 xcircuit
drwxr-xr-x 2 vsduser docker 4096 Jun 28 2021 qflow
drwxr-xr-x 10 vsduser docker 4096 Jun 28 2021 openlane
drwxr-xr-x 2 vsduser docker 4096 Jun 28 2021 ngspice
drwxr-xr-x 2 vsduser docker 4096 Jun 28 2021 netgen
drwxr-xr-x 4 vsduser docker 4096 Jun 28 2021 magic
drwxr-xr-x 2 vsduser docker 4096 Jun 28 2021 klayout
drwxr-xr-x 2 vsduser docker 4096 Jun 28 2021 irsin
vsduser@vdsquadron:~/Desktop/work/tools/openlane_working_dir/pdks/sky130A/libs.tech$ cd ../
vsduser@vdsquadron:~/Desktop/work/tools/openlane_working_dir/pdks/sky130A$ cd libs.ref
vsduser@vdsquadron:~/Desktop/work/tools/openlane_working_dir/pdks/sky130A/libs.ref$ ls -ltr
total 48
drwxr-xr-x 10 vsduser docker 4096 Jun 28 2021 sky130_osu_sc_t18
drwxr-xr-x 12 vsduser docker 4096 Jun 28 2021 sky130_fd_sc_ms
drwxr-xr-x 12 vsduser docker 4096 Jun 28 2021 sky130_fd_sc_ls
drwxr-xr-x 12 vsduser docker 4096 Jun 28 2021 sky130_fd_sc_hs
drwxr-xr-x 12 vsduser docker 4096 Jun 28 2021 sky130_fd_sc_hdll
drwxr-xr-x 8 vsduser docker 4096 Jun 28 2021 sky130_fd_pr
drwxr-xr-x 9 vsduser docker 4096 Jun 28 2021 sky130_sram_macros
drwxr-xr-x 12 vsduser docker 4096 Jun 28 2021 sky130_fd_sc_hvl
drwxr-xr-x 11 vsduser docker 4096 Jun 28 2021 sky130_fd_io
drwxr-xr-x 4 vsduser docker 4096 Jun 28 2021 sky130_nl_xx_hd
drwxr-xr-x 12 vsduser docker 4096 Jun 28 2021 sky130_fd_sc_lp
drwxr-xr-x 12 vsduser docker 4096 Jun 28 2021 sky130_fd_sc_hd
vsduser@vdsquadron:~/Desktop/work/tools/openlane_working_dir/pdks/sky130A/libs.ref$
```

Exploring further the directory sky130_fd_sc_hd in libs.ref and listing its contents we come across technology files like Verilog,spice,techlef,maglef,lib.lef etc. And further exploring lib directory and listing its contents - (speed (ff,ss),temperature(tt)).

Design preparation step

```

vsduser@vsdsquadro... * vsduser@vsdsquadron... * vsduser@vsdsquadron... * vsduser@vsdsquadron... * vsduser@
drwxr-xr-x 15 1000 997 4096 Jun 29 2021 scripts
-rw-r--r-- 1 1000 997 20787 Jun 29 2021 run_designs.py
-rw-r--r-- 1 1000 997 7898 Jun 29 2021 report_generation_wrapper.py
drwxr-xr-x 3 1000 997 4096 Jun 29 2021 regression_results
-rwxr-xr-x 1 1000 997 6519 Jun 29 2021 flow.tcl
drwxr-xr-x 5 1000 997 4096 Jun 29 2021 docs
drwxr-xr-x 5 1000 997 4096 Jun 29 2021 docker_build
drwxr-xr-x 44 1000 997 4096 Jun 29 2021 designs
drwxr-xr-x 2 1000 997 4096 Jun 29 2021 configuration
-rw-r--r-- 1 1000 997 5514 Jun 29 2021 conf.py
-rwxr-xr-x 1 1000 997 966 Jun 29 2021 clean_runs.tcl
-rw-r--r-- 1 1000 997 25509 Jun 29 2021 README.md
-rw-r--r-- 1 1000 997 7273 Jun 29 2021 Makefile
-rw-r--r-- 1 1000 997 11350 Jun 29 2021 LICENSE
-rw-r--r-- 1 1000 997 1285 Jun 29 2021 CONTRIBUTING.md
-rw-r--r-- 1 1000 997 709 Jun 29 2021 AUTHORS.md
-rw-r--r-- 1 1000 1000 963 May 19 2023 default.cvcrc
bash-4.2$ ./flow.tcl -interactive
[INFO]:

```

```

vdsuser@vdsquadron:~$ vdsuser@vdsquadron:~$ vdsuser@vdsquadron:~$ vdsuser@vdsquadron:~$
vdsuser@vdsquadron:~/Desktop/work/tools/openlane_working_dir/openlane$ cd designs/ptcorv32a/
vdsuser@vdsquadron:~/Desktop/work/tools/openlane_working_dir/openlane/designs/ptcorv32a$ ls
total 28
drwxr-xr-x 2 vdsuser docker 4096 Jun 29 2021 src
-rw-r--r-- 1 vdsuser docker 209 Jun 29 2021 sky130A_sky130_fd_sc_ms_config.tcl
-rw-r--r-- 1 vdsuser docker 209 Jun 29 2021 sky130A_sky130_fd_sc_ls_config.tcl
-rw-r--r-- 1 vdsuser docker 209 Jun 29 2021 sky130A_sky130_fd_sc_hs_config.tcl
-rw-r--r-- 1 vdsuser docker 209 Jun 29 2021 sky130A_sky130_fd_sc_hd1l_config.tcl
-rwxr-xr-x 1 vdsuser docker 209 Jun 29 2021 sky130A_sky130_fd_sc_hd_config.tcl
-rwxr-xr-x 1 vdsuser docker 444 Jun 29 2021 config.tcl
vdsuser@vdsquadron:~/Desktop/work/tools/openlane_working_dir/openlane/designs/ptcorv32a$ cd
vdsuser@vdsquadron:~/Desktop/work/tools/openlane_working_dir/openlane$ cd designs/ptcorv32a/
vdsuser@vdsquadron:~/Desktop/work/tools/openlane_working_dir/openlane/designs/ptcorv32a$ ls
total 32
drwxr-xr-x 2 vdsuser docker 4096 Jun 29 2021 src
-rw-r--r-- 1 vdsuser docker 209 Jun 29 2021 sky130A_sky130_fd_sc_ms_config.tcl
-rw-r--r-- 1 vdsuser docker 209 Jun 29 2021 sky130A_sky130_fd_sc_ls_config.tcl
-rw-r--r-- 1 vdsuser docker 209 Jun 29 2021 sky130A_sky130_fd_sc_hs_config.tcl
-rw-r--r-- 1 vdsuser docker 209 Jun 29 2021 sky130A_sky130_fd_sc_hd1l_config.tcl
-rwxr-xr-x 1 vdsuser docker 209 Jun 29 2021 sky130A_sky130_fd_sc_hd_config.tcl
-rwxr-xr-x 1 vdsuser docker 444 Jun 29 2021 config.tcl
drwxr-xr-x 3 vdsuser vdsuser 4096 Jan 30 09:19 runs
vdsuser@vdsquadron:~/Desktop/work/tools/openlane_working_dir/openlane/designs/ptcorv32a$ cd
vdsuser@vdsquadron:~/Desktop/work/tools/openlane_working_dir/openlane/designs/ptcorv32a$ run

```

CHIP DESIGNING COURSE

```
vsduser@vsdsquadron:~/Desktop/work/tools/openlane_working_dir/openlane/designs/picorv32a$ cd runs
vsduser@vsdsquadron:~/Desktop/work/tools/openlane_working_dir/openlane/designs/picorv32a/runs$ ls -ltr
total 4
drwxr-xr-x 6 vsduser vsduser 4096 Jan 30 10:13 30-01_03-49
vsduser@vsdsquadron:~/Desktop/work/tools/openlane_working_dir/openlane/designs/picorv32a/runs$
```

```
drwxr-xr-x 2 vsduser vsduser 4096 Jan 30 09:19 routing
drwxr-xr-x 2 vsduser vsduser 4096 Jan 30 09:19 magic
drwxr-xr-x 2 vsduser vsduser 4096 Jan 30 09:19 lvs
drwxr-xr-x 2 vsduser vsduser 4096 Jan 30 09:19 klayout
drwxr-xr-x 2 vsduser vsduser 4096 Jan 30 09:19 cvc
drwxr-xr-x 2 vsduser vsduser 4096 Jan 30 10:13 synthesis
vsduser@vsdsquadron:~/Desktop/work/tools/openlane_working_dir/openlane/designs/picorv32a/runs/30-01_03-49/results$ cd synthesis
total 1848
lrwxrwxrwx 1 vsduser vsduser 29 Jan 30 09:19 merged_unpadded.lef -> ../../tmp/merged_unpadded.lef
-rw-r--r-- 1 vsduser vsduser 1889131 Jan 30 10:13 picorv32a.synthesis.v
vsduser@vsdsquadron:~/Desktop/work/tools/openlane_working_dir/openlane/designs/picorv32a/runs/30-01_03-49/results/synthesis$ less picorv32a.synthesis.v
vsduser@vsdsquadron:~/Desktop/work/tools/openlane_working_dir/openlane/designs/picorv32a/runs/30-01_03-49/results/synthesis$ cd ../../
total 52
-rwxr-xr-x 1 vsduser vsduser 170 Jun 28 2021 PDK_SOURCES
drwxr-xr-x 11 vsduser vsduser 4096 Jan 30 09:19 results
drwxr-xr-x 11 vsduser vsduser 4096 Jan 30 09:19 reports
-rw-r--r-- 1 vsduser vsduser 15 Jan 30 09:19 OPENLANE_VERSION
drwxr-xr-x 11 vsduser vsduser 4096 Jan 30 09:19 logs
drwxr-xr-x 11 vsduser vsduser 4096 Jan 30 10:10 tmp
drwxr-xr-x 1 vsduser vsduser 2846 Jan 30 10:13 cnds.log
-rw-r--r-- 1 vsduser vsduser 20953 Jan 30 10:13 config.tcl
vsduser@vsdsquadron:~/Desktop/work/tools/openlane_working_dir/openlane/designs/picorv32a/runs/30-01_03-49$
```

Flip flop ratio

CHIP DESIGNING COURSE

sky130_fd_sc_hd_a220_2	1013
sky130_fd_sc_hd_a2bb20_2	1740
sky130_fd_sc_hd_a2bb20t_2	81
sky130_fd_sc_hd_a310_2	2
sky130_fd_sc_hd_a310t_2	49
sky130_fd_sc_hd_a310t_2	7
sky130_fd_sc_hd_a320_2	46
sky130_fd_sc_hd_a410_2	1
sky130_fd_sc_hd_and2_2	157
sky130_fd_sc_hd_and3_2	58
sky130_fd_sc_hd_and4_2	345
sky130_fd_sc_hd_and4b_2	1
sky130_fd_sc_hd_buf_1	1656
sky130_fd_sc_hd_buf_2	8
sky130_fd_sc_hd_comb_1	42
sky130_fd_sc_hd_dfxt_2	1613
sky130_fd_sc_hd_inv_2	1615
sky130_fd_sc_hd_mux2_1	1224
sky130_fd_sc_hd_mux2_2	2
sky130_fd_sc_hd_mux4_1	221

Number of wires:	14596
Number of wire bits:	14978
Number of public wires:	1565
Number of public wire bits:	1947
Number of memories:	0
Number of memory bits:	0
Number of processes:	0
Number of cells:	14876
sky130_fd_sc_hd_a21110_2	1
sky130_fd_sc_hd_a2110_2	35
sky130_fd_sc_hd_a2110t_2	60
sky130_fd_sc_hd_a21b0_2	149
sky130_fd_sc_hd_a21b0t_2	8
sky130_fd_sc_hd_a210_2	57
sky130_fd_sc_hd_a210t_2	244
sky130_fd_sc_hd_a2210_2	86
sky130_fd_sc_hd_a220_2	1613

FLIP FLOP RATIO = NO OF DFFs / NO OF CELLS * 100

= 1613/14876*100

=10.84

```

vdsuser@vdsquadron:~/Desktop/work/tools/openlane_working_dir/openlane/designs/picorv32a/runs/30-01_03-49$ cd results/synthesis
vdsuser@vdsquadron:~/Desktop/work/tools/openlane_working_dir/openlane/designs/picorv32a/runs/30-01_03-49/results/synthesis$ ls -ltr
total 1848
lrwxrwxrwx 1 vdsuser vdsuser 29 Jan 30 09:19 merged_unpadded.lef -> ../../tmp/merged_unpadded.lef
-rw-r--r-- 1 vdsuser vdsuser 1889131 Jan 30 10:13 picorv32a.synthesis.v
vdsuser@vdsquadron:~/Desktop/work/tools/openlane_working_dir/openlane/designs/picorv32a/runs/30-01_03-49/results/synthesis$

```